```
!pip uninstall -y langchain langchain-core langchain-community langchain-huggingface
!pip install -q \
    langchain \
    langchain-core \
    langchain-community \
    langchain-huggingface \
    faiss-cpu \
    pypdf \
    sentence-transformers \
    gradio
```

```
Found existing installation: langchain 1.2.12
Uninstalling langchain-1.2.12:
  Successfully uninstalled langchain-1.2.12
Found existing installation: langchain-core 1.2.19
Uninstalling langchain-core-1.2.19:
  Successfully uninstalled langchain-core-1.2.19
WARNING: Skipping langchain-community as it is not installed.
WARNING: Skipping langchain-huggingface as it is not installed.
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 112.5/112.5 kB 12
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 506.7/506.7 kB 40
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.5/2.5 MB 104.7
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 23.8/23.8 MB 88.0
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 333.7/333.7 kB 28
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.0/1.0 MB 72.0 M
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 65.0/65.0 kB 7.4
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 51.0/51.0 kB 5.2
ERROR: pip's dependency resolver does not currently take into account all the packages t
google-colab 1.0.0 requires requests==2.32.4, but you have requests 2.33.0 which is inco
```

```
import os
os.kill(os.getpid(), 9)
```

```
!pip install -q transformers --upgrade
```

```
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 10.1/10.1 MB 76.5
```

```
import os
os.kill(os.getpid(), 9)
```

```
from langchain_community.document_loaders import PyPDFLoader
from langchain_text_splitters import RecursiveCharacterTextSplitter

loader = PyPDFLoader("14.pdf")
documents = loader.load()
print(f"✅ عدد الصفحات: {len(documents)}")

text_splitter = RecursiveCharacterTextSplitter(
    chunk_size=600,
    chunk_overlap=100
)
chunks = text_splitter.split_documents(documents)
print(f"✅ عدد القطع: {len(chunks)}")
```

```
✅ عدد الصفحات: 289
✅ عدد القطع: 1108
```

```
from langchain_huggingface import HuggingFaceEmbeddings
from langchain_community.vectorstores import FAISS

embeddings = HuggingFaceEmbeddings(
    model_name="sentence-transformers/all-MiniLM-L6-v2"
```

```
)

vector_db = FAISS.from_documents(chunks, embeddings)
vector_db.save_local("faiss_index")
print("✅ تم إنشاء قاعدة البيانات بنجاح")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public model
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to
modules.json: 100%  349/349 [00:00<00:00, 34.9kB/s]
config_sentence_transformers.json: 100%  116/116 [00:00<00:00, 10.5kB/s]
README.md: 10.5k/? [00:00<00:00, 879kB/s]
sentence_bert_config.json: 100%  53.0/53.0 [00:00<00:00, 6.09kB/s]
config.json: 100%  612/612 [00:00<00:00, 71.6kB/s]
model.safetensors: 100%  90.9M/90.9M [00:01<00:00, 457MB/s]
Loading weights: 100%  103/103 [00:00<00:00, 5798.84it/s]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:can be ignored when loading from different task/architecture; not ok if you
tokenizer_config.json: 100%  350/350 [00:00<00:00, 30.7kB/s]
vocab.txt: 232k/? [00:00<00:00, 10.9MB/s]
tokenizer.json: 466k/? [00:00<00:00, 21.4MB/s]
special_tokens_map.json: 100%  112/112 [00:00<00:00, 12.1kB/s]
config.json: 100%  190/190 [00:00<00:00, 16.0kB/s]
✅ تم إنشاء قاعدة البيانات بنجاح
```

```
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM, pipeline
from langchain_huggingface import HuggingFacePipeline

model_id = "microsoft/phi-2"
print("⏳ جارٍ تحميل النموذج...")

tokenizer = AutoTokenizer.from_pretrained(
    model_id,
    trust_remote_code=True
)

model = AutoModelForCausalLM.from_pretrained(
    model_id,
    torch_dtype=torch.float16,
    device_map="auto",
    trust_remote_code=True
)

phi_pipeline = pipeline(
    "text-generation",
    model=model,
```

```
    tokenizer=tokenizer,
    max_new_tokens=256,
    temperature=0.1,
    do_sample=True,
    repetition_penalty=1.1
)

llm = HuggingFacePipeline(pipeline=phi_pipeline)
print("✅ تم تحميل Phi-2 بنجاح")
```

```
⏳ جارٍ تحميل النموذج...
config.json: 100%                                                    735/735 [00:00<00:00, 66.6kB/s]

tokenizer_config.json:        7.34k/? [00:00<00:00, 703kB/s]

tokenizer.json:        2.11M/? [00:00<00:00, 61.8MB/s]

added_tokens.json:        1.08k/? [00:00<00:00, 105kB/s]

special_tokens_map.json: 100%                                                99.0/99.0 [00:00<00:00, 8.40kB/s]
`torch_dtype` is deprecated! Use `dtype` instead!
model.safetensors.index.json:        35.7k/? [00:00<00:00, 3.15MB/s]

Download complete: 100%        5.56G/5.56G [01:05<00:00, 78.9MB/s]

Fetching 2 files: 100%                                               2/2 [01:05<00:00, 65.29s/it]

Loading weights: 100%                                               453/453 [00:18<00:00, 24.90it/s]

generation_config.json: 100%                                         124/124 [00:00<00:00, 10.5kB/s]
Passing `generation_config` together with generation-related arguments=({'do_sample', 't
✅ تم تحميل Phi-2 بنجاح
```

```
from langchain_core.prompts import PromptTemplate
from langchain_core.runnables import RunnablePassthrough
from langchain_core.output_parsers import StrOutputParser

template = """<s>[INST]
أنت مساعد ذكي متخصص في لوائح جامعة فرات.
استخدم فقط الفقرات التالية للإجابة على سؤال المستخدم.
إذا لم تجد الإجابة، قل بأدب أنك لا تعرف.

السياق:
{context}

السؤال: {question}
[/INST]
</s>"""

QA_PROMPT = PromptTemplate.from_template(template)
retriever = vector_db.as_retriever(search_kwargs={"k": 3})

def format_docs(docs):
    return "\n\n".join(doc.page_content for doc in docs)

rag_chain = (
    {"context": retriever | format_docs, "question": RunnablePassthrough()}
    | QA_PROMPT
    | llm
    | StrOutputParser()
)

print("✅ تم بناء سلسلة RAG بنجاح")
```

```
✅ تم بناء سلسلة RAG بنجاح
```

```
import gradio as gr

def ask_mistral_gpt(question, history):
    try:
        answer = rag_chain.invoke(question)
        if "[/INST]" in answer:
            answer = answer.split("[/INST]")[-1].strip()
        return answer
    except Exception as e:
        return f"⚠️ حدث خطأ: {str(e)}"

demo = gr.ChatInterface(
    fn=ask_mistral_gpt,
    title="🎓 Fırat-GPT — Mistral RAG",
    description="اسأل عن أي شيء يخص لوائح جامعة فرات.",
    theme="soft",
    examples=[
        "ما هو نموذج Form-30؟",
        "كم عدد أعضاء لجنة تحكيم الماجستير؟",
        "ما هي شروط القبول في الدراسات العليا؟"
    ]
)

demo.launch(share=True)
```

```
/usr/local/lib/python3.12/dist-packages/gradio/chat_interface.py:347: UserWarning: The '
  self.chatbot = Chatbot(
Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: https://6fcea8f6b0c54dd492.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gra
```

gradio

No interface is running right now

```
!nvidia-smi
```

```
/bin/bash: line 1: nvidia-smi: command not found
```